J. Nuclear Energy, 1956, Vol. 2, pp. 153 to 167. Pergamon Press Ltd., London

EFFECT OF MODERATOR TEMPERATURE UPON NEUTRON FLUX IN INFINITE, CAPTURING MEDIUM

R. R. COVEYOU, R. R. BATE, and R. K. OSBORN
Oak Ridge National Laboratory, Oak Ridge, Tennessee

(*Received* 5 *September* 1955)

Abstract—The integral equation governing the dependence upon speed of the neutron flux for moderation in an infinite, capturing medium has been solved by Monte Carlo methods. Effects on the flux due to the thermal motions of the scattering and absorbing atoms are taken into account within the context of the assumption that the thermal speed distributions are Maxwell-Boltzmann in character. Results for several different scattering masses and several different $(1/v)$-absorption cross-sections are presented in detail.

1. INTRODUCTION

THE slowing down of neutrons in a capturing medium, taking into account the velocity distribution of the moderating and absorbing atoms, has been investigated by several authors. WIGNER and WILKINS (1944) obtained numerical solutions of the exact integral equation for the case of an infinite medium containing a homogeneous mixture of gaseous hydrogen and absorbing atoms characterized by a $(1/v)$-absorption cross-section. Their method of solution is not rigorously applicable to cases for which the moderator mass is not equal to the neutron mass, although WILKINS (1948) has developed an approximate application of the method.† JAFFÉ (1952) exhibited the neutron transport equation in the form

$$\boldsymbol{v} \cdot \nabla n(\boldsymbol{r},\boldsymbol{v}) + (\textstyle\sum_a + \sum_s) v n(\boldsymbol{r},\boldsymbol{v}) = S(\boldsymbol{r},\boldsymbol{v}) + \textstyle\sum_s \int G(\boldsymbol{v},\boldsymbol{v}') v' n(\boldsymbol{r},\boldsymbol{v}') d^3 v' \qquad (1)$$

with an explicit determination of the scattering function G, and obtained approximate solutions for certain special cases by iteration. The particular virtue of this formulation of the problem is that it still contains angle as well as energy dependence and provides a feasible starting-point for the development of an "age theory" for finite systems with temperature modifications. Of course, in an infinite medium and for gaseous moderation, equation (1) reduces to one derived by WIGNER and WILKINS (1944).

NIESTLIE and BERMAN (1954) solved equation (1) for the infinite medium case with Be moderation by numerical integration.‡ The same equation was also solved by BROWN and ST. JOHN (1954) in a similar fashion for D_2O moderation. However, the latter investigators chose a scattering function G which took into account the principal effects on the scattering due to rotational transitions and the binding of the D_2O molecule, and, consequently, is somewhat modified relative to the scattering function (WIGNER and WILKINS) obtained on the assumption of neutrons scattering from free gas atoms in a Maxwell-Boltzmann distribution.

† Some results of calculations employing this method were generously supplied to us by J. R. TRIPLETT, General Electric Co., Richland, Washington, for purposes of comparison with the Monte Carlo calculations. It was found that, even in regions of the parameters in which the WILKINS's asymptotic differential equation was not expected to be valid, the results were in close agreement with those described herein.

‡ The numerical integration scheme developed by these investigators is very fast. We are grateful to them for supplying us with the results of several calculations prior to publication. Their results were compared with the calculations described herein and the agreement was excellent.

The above methods consist of either replacing the integral equation by an equivalent differential equation (exact for hydrogen scattering and approximate otherwise) and then obtaining numerical solutions of the differential equation, or by direct numerical (or approximate) solution of the integral equation itself. Because of the availability of a high-speed computer and because of its unique capacity for performing an enormous number of simple operations in a relatively short time, we have chosen to attack the problem by the Monte Carlo method.

2. DETAILS OF THE METHOD OF CALCULATION

We have restricted our attention exclusively to the energy (or speed) spectrum of neutrons slowing down in an infinite, homogeneous medium. We further assume that the moderating and absorbing atoms are unbound and distributed energy-wise according to a Maxwell-Boltzmann distribution. We see from the theoretical development below that the technique is readily applicable to any mixture of systems provided that all scattering cross sections are independent of relative speed and that all *effective* absorption cross-sections are known.

The crux of the method is readily outlined. A neutron is introduced into the system with a pre-assigned source-speed. A decision is then made as to the nature of the first event to be experienced by the neutron; i.e., is it to be absorbed or scattered, and by what species respectively. If the event is an absorption, a new neutron is introduced at the source-speed and repeat. If the event is to be a scattering, a decision has to be made regarding the speed of the target particle and the relative angle of approach of the neutron to the target. Having made this decision, it is then a simple matter, on the assumption of isotropic scattering in the centre of mass, to select the new speed of the neutron. This new neutron now repeats the same sequence of decisions as the original one until it is finally absorbed, thus terminating a history.

A closer scrutiny indicates, however, that a Monte Carlo calculation of the slowing-down process cannot proceed quite so directly as the above outline suggests. In the first place, the decision as to whether a given event shall be an absorption or a scattering depends, for a neutron of speed v, upon the ratio

$$\sum_{ae}(v)/\sum_{se}(v)$$

where the cross-sections indicated represent the total, macroscopic, *effective* cross-sections for absorption and scattering, respectively. The effective cross-section of given character for the ith species is defined as

$$\sum_e^i(v) = N_i \left(\frac{\alpha_i}{\pi}\right)^{3/2} \int \sigma^i(|\boldsymbol{v} - \boldsymbol{V}|) \frac{|\boldsymbol{v} - \boldsymbol{V}|}{v} e^{-\alpha_i V^2} d^3V \tag{2}$$

where $\alpha_i = M_i/2kT$; on the assumption that the target particles are in a Maxwell-Boltzmann distribution. The σ^i is the microscopic cross-section for an event of a given kind, and is indicated as being in general a function of the relative speed of the neutron to target. The factor, N_i, is the concentration of the ith species.

It is to be noted that the effective cross-section for a $(1/v)$-absorber for which the microscopic cross-section is

$$\sigma = B/|\boldsymbol{v} - \boldsymbol{V}|$$

is simply

$$\sum_{ea}(v) = NB/v. \tag{3}$$

We have assumed throughout that the microscopic scattering cross-sections are constants; hence the effective scattering cross-section is

$$\Sigma_{es}^{i}(v) = N_i\,\sigma_s^{\,i}\left[\frac{e^{-\alpha_i v^2}}{\sqrt{\pi\alpha_i}\,v} + \left(1 + \frac{1}{2\alpha_i v^2}\right)\operatorname{erf}(\sqrt{\alpha_i}\,v)\right] \tag{4}$$

where

$$\operatorname{erf}(X) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-y^2}\,dy$$

and the $\sigma_s^{\,i}$ are constants. It is both undesirable and unnecessary to compute the Σ_{es}^{i} at every stage of the process, but this avoidance occasions some indirectness in the method of calculation. A more fundamental and complicating consideration arises, however, in connection with the choice of the parameters characterizing a scattering—speed of target and direction of motion of neutron to target—subsequent to the decision that the event is indeed a scattering. This choice must be made by an appropriate selection from the speed and angle distribution of targets as seen by the neutron

$$P(V, \mu \mid v)\,dV\,d\mu = \frac{\sum_i N_i\left(\frac{\alpha_i}{\pi}\right)^{3/2}\sigma_s^{\,i}\,|\,\boldsymbol{v} - \boldsymbol{V}\,|\,V^2 e^{-\alpha_i V^2}\,2\pi\,dV\,d\mu}{\sum_i \Sigma_{es}^{i}(v)}\,. \tag{5}$$

The presence of the relative speed in the distribution function appears to provide a considerable barrier to any direct method of selection.

The technique whereby these complications were satisfactorily circumvented† will be outlined below, using as an example the case of a single absorber with arbitrary absorption cross-section and a single scatterer with constant cross-section. (The scattering and absorbing species are not necessarily the same.) The generalization to homogeneous mixtures of arbitrary composition is straightforward.

We express the reaction rate for neutrons of speed v with atoms whose speeds lie in dV about V and whose directions relative to $\boldsymbol{v}$ lie in $d\mu$ about μ (μ being the cosine of the angle of approach) as

$$\begin{aligned} R(V, \mu \mid v)\,dV\,d\mu = {} & \left[4\pi N_s\left(\frac{\alpha_s}{\pi}\right)^{3/2}\sigma_s v V^2 e^{-\alpha_s V^2}\,dV\,\frac{d\mu}{2}\right. \\ & \left. + 4\pi N_s\left(\frac{\alpha_s}{\pi}\right)^{3/2}\sigma_s V^3 e^{-\alpha_s V^2}\,dV\,\frac{d\mu}{2}\right]\frac{|\boldsymbol{v} - \boldsymbol{V}|}{v + V} \\ & + 4\pi N_a\left(\frac{\alpha_a}{\pi}\right)^{3/2}\sigma_a\,|\,\boldsymbol{v} - \boldsymbol{V}\,|\,V^2 e^{-\alpha_a V^2}\,dV\,\frac{d\mu}{2}\,. \end{aligned} \tag{6}$$

Now choose three numbers (P_1, P_2, P_3) such that

$$P_1 \propto v\,\Sigma_{ae}(v)$$

$$P_2 \propto 4\pi N_s\left(\frac{\alpha_s}{\pi}\right)^{3/2}\sigma_s\,\frac{v}{C_2}$$

$$P_3 \propto 4\pi N_s\left(\frac{\alpha_s}{\pi}\right)^{3/2}\sigma_s\,\frac{1}{C_3}$$

† This method was suggested to us by H. KAHN of the Rand Corporation in a private communication.

where $P_1 + P_2 + P_3 = 1$. The numbers C_2 and C_3 are defined to be

$$C_2 = \left(\int_0^\infty V^2 e^{-\alpha_s V^2}\, dV\right)^{-1} = \frac{4\alpha_s^{3/2}}{\sqrt{\pi}}$$

$$C_3 = \left(\int_0^\infty V^3 e^{-\alpha_s V^2}\, dV\right)^{-1} = 2\alpha_s^2.$$

Thus

$$P_1 = v\textstyle\sum_{ae}(v)/D(v), \tag{6a}$$

$$P_2 = N_s\sigma_s v/D(v), \tag{6b}$$

$$P_3 = (2N_s\sigma_s/\sqrt{\pi\alpha_s})/D(v), \tag{6c}$$

where

$$D(v) = v\textstyle\sum_{ae}(v) + N_s\sigma_s v + 2N_s\sigma_s/\sqrt{\pi\alpha_s}.$$

Assume that a neutron is introduced into the medium with preassigned source speed v. Choose a number η^* from a uniform distribution on the interval (0, 1) and test whether

$$P_1 \lesseqgtr \eta^*.$$

If greater, record an absorption at speed v and repeat with a new neutron. If less, test whether

$$P_1 + P_2 \gtreqless \eta^*.$$

If greater, choose V from the distribution

$$C_2 V^2 e^{-\alpha_s V^2}\, dV, (0, \infty)$$

and if less, choose V from

$$C_3 V^3 e^{-\alpha_s V^2}\, dV, (0, \infty).$$

In either case, choose μ from the distribution

$$\frac{d\mu}{2}, (-1, +1).$$

Compute $|\boldsymbol{v} - \boldsymbol{V}| = \sqrt{v^2 + V^2 - 2vV\mu}$ and choose η from $d\eta$, (0, 1) and compare

$$\frac{|\boldsymbol{v} - \boldsymbol{V}|}{v + V} \gtreqless \eta.$$

If less, then these scattering parameters are to be rejected and the whole process is to be repeated, including the determination as to whether the event is a scattering or absorption. However, if greater, then a scattering is recorded at this speed. For any collision, one has

$$\gamma\boldsymbol{v} + \beta\boldsymbol{V} = \gamma\boldsymbol{v}' + \beta\boldsymbol{V}' = \boldsymbol{g}$$

where

$$\gamma = \frac{m}{m + M} = \frac{1}{A + 1}, \qquad \beta = \frac{M}{m + M} = \frac{A}{A + 1}$$

and g = velocity of centre of mass, the primed and unprimed quantities representing post- and pre-collision variables, respectively. Since for elastic collisions

$$|\boldsymbol{v} - \boldsymbol{V}| = |\boldsymbol{v}' - \boldsymbol{V}'|,$$

the new speed of the neutron may be immediately expressed as

$$|\boldsymbol{v}'| = |\beta(\boldsymbol{v}' - \boldsymbol{V}') + \boldsymbol{g}| = \sqrt{\beta^2 |\boldsymbol{v} - \boldsymbol{V}|^2 + g^2 + 2\beta |\boldsymbol{v} - \boldsymbol{V}| g\lambda} \tag{7}$$

where λ is the cosine of the scattering angle in the centre of mass. The assumption of isotropic scattering allows one to choose λ from

$$\frac{d\lambda}{2}, (-1, +1).$$

The neutron's new speed is now determined and the above process is repeated with the replacement of v' for v. The repetition continues to ultimate absorption, which terminates a history.

It is clear that the above technique provides scattering parameters from the appropriate relative distribution; but it is perhaps not quite so clear that the decision as to whether an event is an absorption or a scattering is correctly made. To establish this, consider the probability that an event is an absorption, P_a. Then if P_A is the probability of accepting a given set of scattering parameters in a single pass; i.e., no rejection on the test $\dfrac{|\boldsymbol{v} - \boldsymbol{V}|}{v + V} \gtrless \eta$,

$$\begin{aligned} P_a &= P_1 + (1 - P_1)(1 - P_A)P_1 + (1 - P_1)(1 - P_A)P_1(1 - P_1)(1 - P_A)P_1 + \cdots \\ &= P_1 \sum_{n=0}^{\infty} [(1 - P_1)(1 - P_A)]^n = \frac{P_1}{1 - (1 - P_1)(1 - P_A)}. \end{aligned} \tag{8}$$

The probability that the event is a scattering, P_S, is simply

$$P_S = 1 - P_a = \frac{(1 - P_1)P_A}{1 - (1 - P_1)(1 - P_A)}. \tag{9}$$

Thus the relative probability of absorption to scattering is

$$\frac{P_a}{P_s} = \frac{P_1}{(1 - P_1)P_A} = \frac{P_1}{(P_2 + P_3)P_A}. \tag{10}$$

But P_A, the probability of accepting a given set of scattering parameters in a single pass, is

$$\begin{aligned} P_A &= \int_{V,\mu} \frac{P_2}{P_2 + P_3} C_2 V^2 e^{-\alpha_s V^2} dV \frac{d\mu}{2} \frac{|\boldsymbol{v} - \boldsymbol{V}|}{v + V} \\ &\quad + \int_{V,\mu} \frac{P_3}{P_2 + P_3} C_3 V^3 e^{-\alpha_s V^2} dV \frac{d\mu}{2} \frac{|\boldsymbol{v} - \boldsymbol{V}|}{v + V} \\ &= \frac{4\pi N_s \sigma_s \left(\dfrac{\alpha_s}{\pi}\right)^{3/2}}{(P_2 + P_3)\, D(v)} \frac{1}{2} \int_{V,\mu} |\boldsymbol{v} - \boldsymbol{V}| e^{-\alpha_s V^2} V^2 dV\, d\mu \\ &= \frac{v \sum_{es}(v)}{(P_2 + P_3)\, D(v)}. \end{aligned} \tag{11}$$

Therefore

$$\frac{P_a}{P_s} = \frac{\sum_{ae}(v)/D(v)}{(P_2 + P_3)\dfrac{v\sum_{se}(v)}{(P_2 + P_3)D(v)}} = \frac{\sum_{ae}(v)}{\sum_{se}(v)}$$

as is required.

It is to be noted that the formulae for the numbers (P_1, P_2, P_3) take on a particularly simple form if the absorption cross-section is $(1/v)$; i.e.,

$$\sigma_a = B/|\boldsymbol{v} - \boldsymbol{V}|,$$

and hence

$$\sum_{ae}(v) = N_a B/v;$$

so that

$$P_1 = \frac{N_a B}{D^*(v)}, \tag{12a}$$

$$P_2 = \frac{N_s \sigma_s v}{D^*(v)}, \tag{12b}$$

$$P_3 = \frac{2N_s\sigma_s/\sqrt{\pi\alpha_s}}{D^*(v)} \tag{12c}$$

where

$$D^*(v) = N_a B + N_s\sigma_s v + 2N_s\sigma_s/\sqrt{\pi\alpha_s}$$

3. DISCUSSION OF RESULTS

It is convenient, both for purposes of calculation as well as for presentation of results, to introduce the dimensionless variable

$$y = v/v_T, \qquad v_T = \sqrt{\frac{3kT}{m}},$$

where m is neutron mass. Note that this choice of normalizing speed, v_T, corresponds to the speed at the peak of the flux rather than number-density for neutrons in a Maxwell-Boltzmann distribution. When the formulae for the P's (in the case of $(1/v)$ absorption) and for the distributions are expressed in terms of the variable y, it is seen that the cross-sections and moderator temperature enter into the calculations explicitly only in terms of a parameter

$$K = \frac{N_a B}{N_s\sigma_s v_t} = \sqrt{\frac{2}{3}}\sqrt{\frac{293}{T_m}}\frac{N_a}{N_s}\frac{\sigma_a\,(@\ 2200\ \text{m/s})}{\sigma_s}.$$

The only other parameter that appears explicitly is the ratio of the mass of the scatterer to that of the neutron,

$$A = \frac{M_s}{m}.$$

The results of a calculation are obtained from the machine in the form of a histogram of the number of scattering events and the number of absorption events which occur in each of a set of intervals into which the range of the independent

variable has been subdivided. A single calculation consisted of as many histories as might be required to provide a total of at least 65,000 scattering events distributed over the whole range. The time for a calculation was of the order of four hours. The range of speeds was taken to be from 0 to $32v_T$ and was subdivided into 512 equal intervals (a width in energy of about 0·002 eV at $T = 293°K$).

The number of events of a given kind is, of course, just the collision density for collisions of that kind. As, in general, there were many more scatterings per interval than absorptions, the flux at the end of an interval was obtained from the scattering density according to the relation

$$\phi(y_i) = \frac{N_s(y_i) + N_s(y_{i+1})}{2N_H \Sigma_{es}(y_i)},$$

where Σ_{es} is given by equation (4). N_H is the number of histories, and is introduced to provide the normalization

$$\int \Sigma_{ae}(y)\phi(y)\, dy = 1.$$

The results of calculations for several different values of the parameters K and A are summarized in Figs. 1–11. In each case, three functions have been plotted versus the independent variable $X = 16y$ ($X = 16$ corresponds, therefore, to

$$v = v_T):$$

(1) The computed flux characterized by the high-speed $(1/v)$-tail;

(2) The number density defined in terms of the computed flux according to $N(X) = \phi(X)/X$;

(3) The Maxwell-Boltzmann flux that would obtain if the neutron source were itself proportional to the number density. Thus in the thermal range, the distortion of (1) relative to (3) is a measure of the effect of the $(1/v)$-absorber on the neutron-speed distribution.

The two cases for $A = 1$, $K = 0·18$ and 0·36 are duplications of calculations performed by WIGNER and WILKINS. Good agreement with their results was obtained, as is revealed in Fig. 12, in which the WIGNER-WILKINS flux taken from reference (1) is plotted alongside the Monte Carlo flux for $K = 0·18$. Furthermore, calculations were performed for cases characterized by $AK \lessapprox 0·01$, and the expected Maxwell-Boltzmann distribution was reproduced in the thermal range, as shown in Fig. 2. Also included explicitly on some of the figures are high-energy points represented by squares calculated from the appropriately normalized $(1/v)$-law, and it is seen that the straight line through these points is a reasonable fit to the computed data in this region. Finally, though the high-energy end of the spectrum is the region of poorest data statistically, the characteristic discontinuities in the flux below a point source are clearly discernible. Figure 13 shows the first discontinuity in the flux for neutrons slowing down from a point source in Be^9.

A useful measure of the flux distortion is provided by the concept of the neutron effective temperature, which is considered here to be that temperature determined by a least-squares fit of a Maxwell-Boltzmann neutron flux to the computed flux in the region

$$O \leq X \leq 35.$$

No significance can be attached to the particular choice of upper limit to the fitting range employed here other than the exclusion of the $(1/v)$-region of the flux. The ratio of the effective neutron temperature to the moderator temperature (T_e/T_m) for each case is included in the corresponding figure, with the exception of the cases $A = 9$, $k = 2^{-3}$ and 1 (Figs. 6 and 7, respectively), which are characterized by such severe flux distortion as to render the concept of an effective neutron temperature practically meaningless.

It is found that the linear relation

$$\frac{T_m}{T_e} = 1 + (1{\cdot}11)AK \tag{13}$$

correlates all of the cases presented here to within ~5 per cent. Such a relation, of course, provides a crude but useful interpolation device for estimating the effective neutron temperature in a particular reactor. For example, in a graphite-moderated, natural uranium fuel reactor one has

$$\frac{T_e}{T_m} = 1 + (1{\cdot}11)A\sqrt{\frac{2}{3}}\sqrt{\frac{293}{T_m}}\frac{\sigma_a}{\sigma_s}\frac{\mu(c)}{\mu(u)}\frac{m(u)}{m(c)} \tag{14}$$

where μ is the gm-atomic-wt, and m is the total mass of the given element in the reactor. Hence, for this example

$$\frac{T_e}{T_m} = 1 + 0{\cdot}89\sqrt{\frac{293}{T_m}}\frac{m(u)}{m(c)}. \tag{15}$$

The neutron spectrum of the Harwell reactor was measured by TAYLOR (1952), and the neutron-effective temperature was computed to be 326°K for a moderator temperature of 325°K. Taking for the mass ratio of this reactor the value

$$\frac{m(u)}{m(c)} \sim \frac{1}{9},$$

and employing the above interpolation relation, one computes

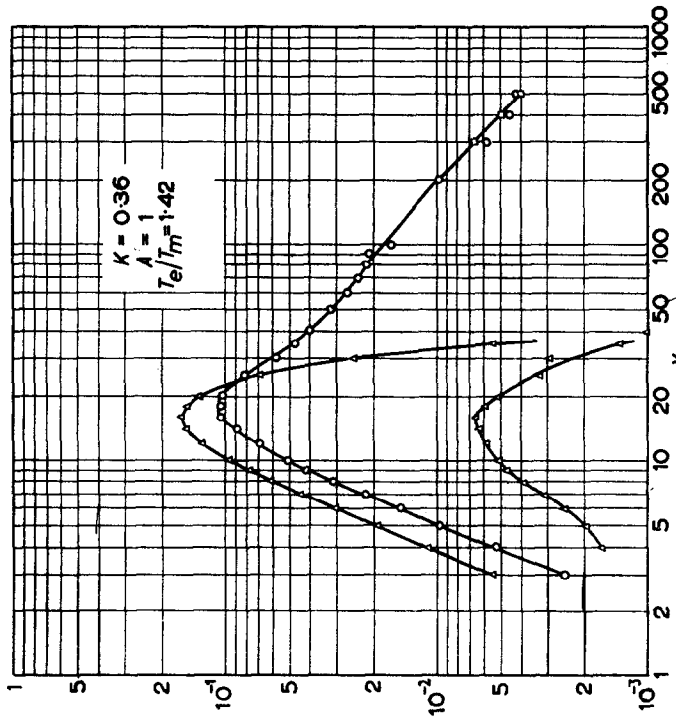
$$\frac{T_e}{T_m} \cong 1 + 0{\cdot}1;\ T_e \sim 355°\text{K}.$$

Considering that the interpolation formula is hardly reliable to better than 5 per cent, and that a finite, inhomogeneous system has been replaced by an infinite, homogeneous medium, the disagreement between the estimate and the measurement is not felt to be too serious.

Calculations taking into account non-$(1/v)$ absorbers and mixtures of both moderators and absorbers are in progress.

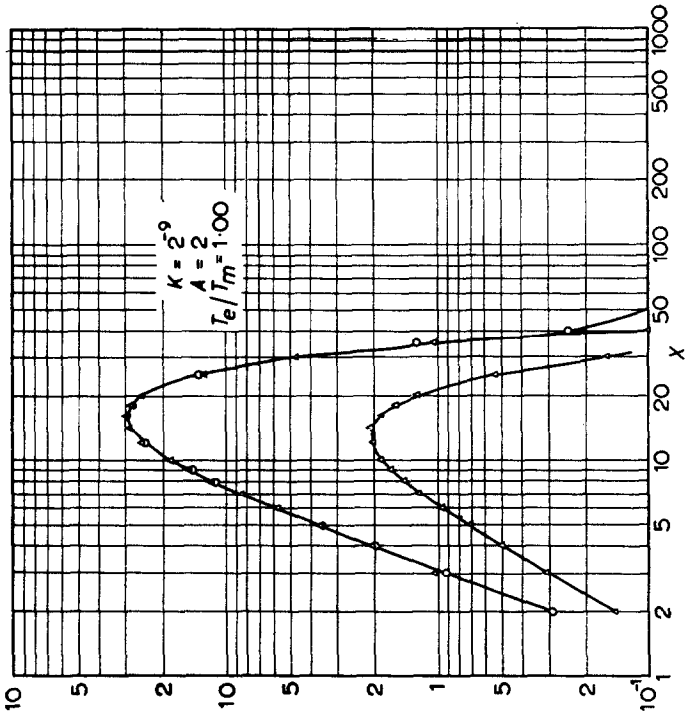
REFERENCES

BROWN, H. D. and ST. JOHN, D. S., Savannah River Operations. Unpublished.
JAFFÉ, G. (1952) *Phys. Rev.*, **88**, 603
NIESTLIE, J. W. JUN. and BERMANN, A. I., Pratt and Whitney Aircraft. Unpublished.
TAYLOR, B. T. (1952), AERE-N/R-1005.
WIGNER, E. P. and WILKINS, J. E., JUN., AECD-2275.
WILKINS, J. E. JUN. (1948) *Annals of Mathematics*, **49**, No. 1.



○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
△ (lower curve) actual number density = (actual flux)/X
□ (1/v) flux with same normalization

FIG. 1.



○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
△ (lower curve) actual number density = (actual flux)/X

FIG. 2.

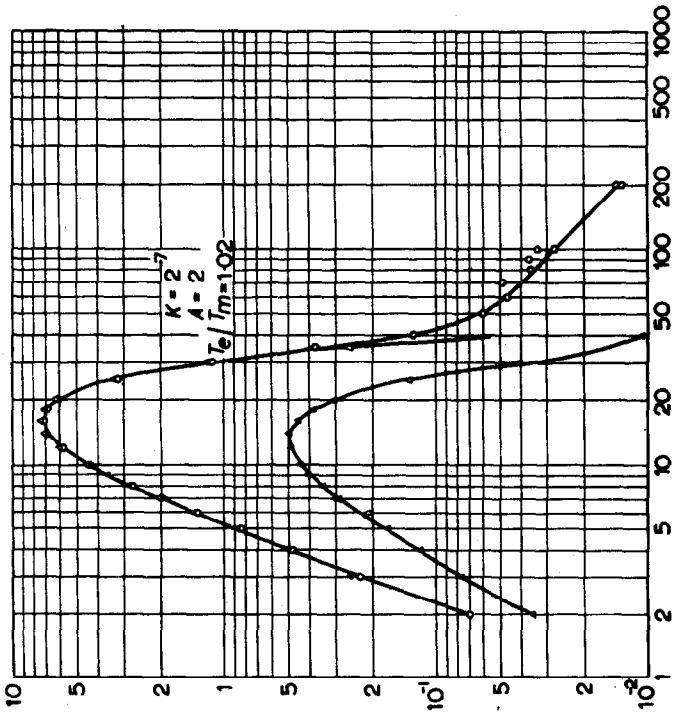


FIG. 3.

○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
◁ (lower curve) actual number density = (actual flux)/X
□ $(1/v)$ flux with same normalization

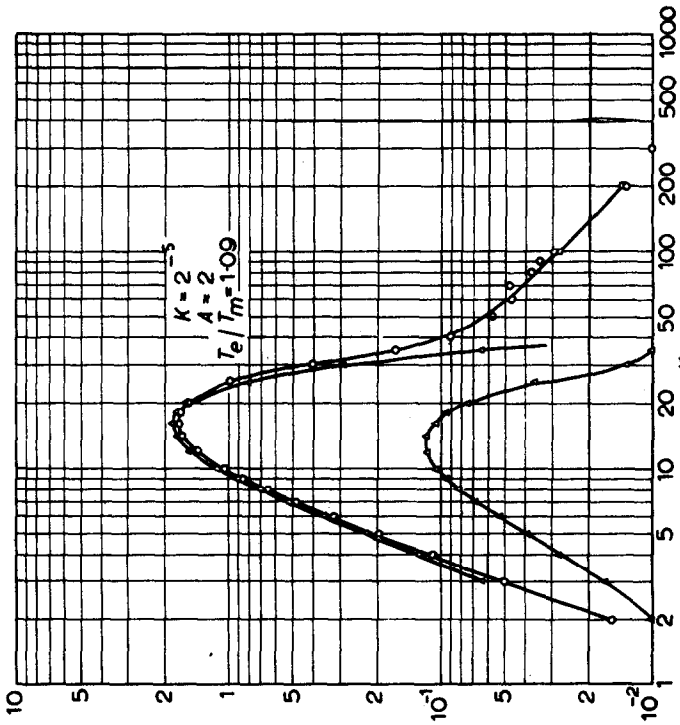


FIG. 4.

○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
◁ (lower curve) actual number density = (actual flux)/X
□ $(1/v)$ flux with same normalization

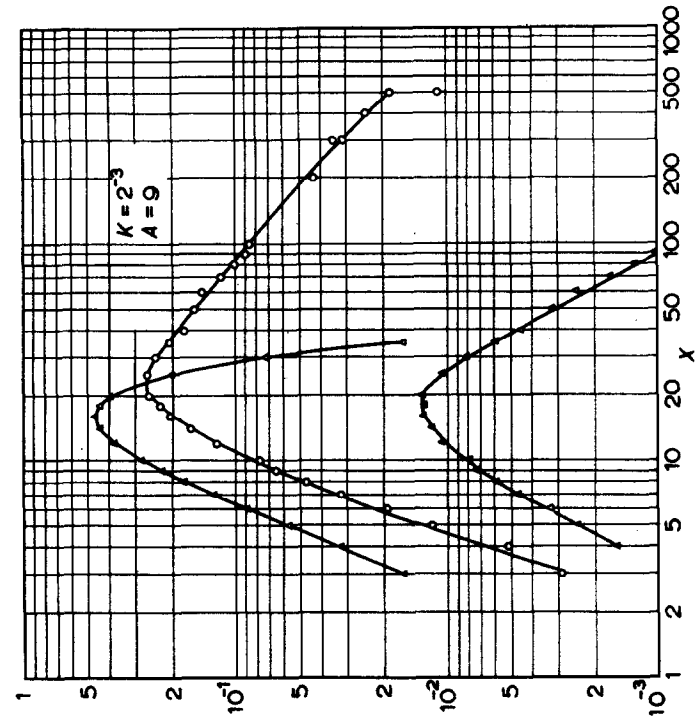


FIG. 6.

○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
△ (lower curve) actual number density = (actual flux)/X

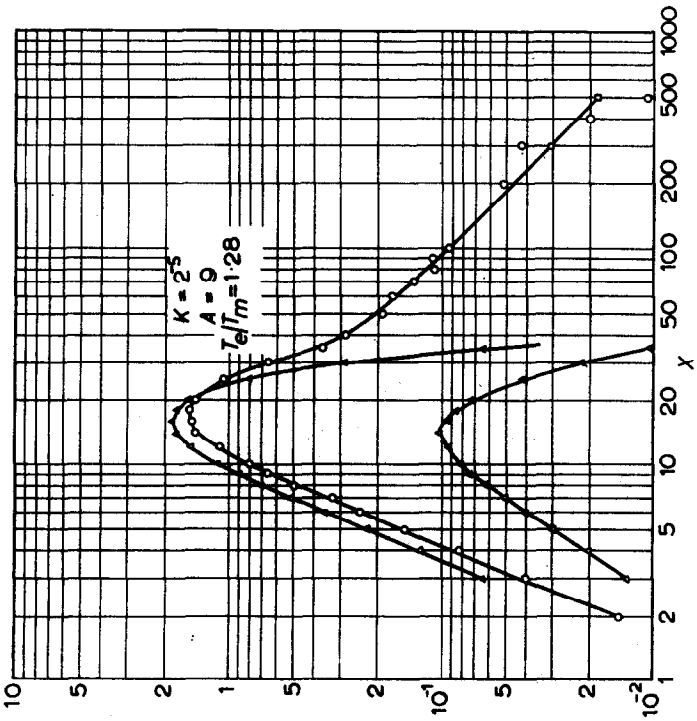
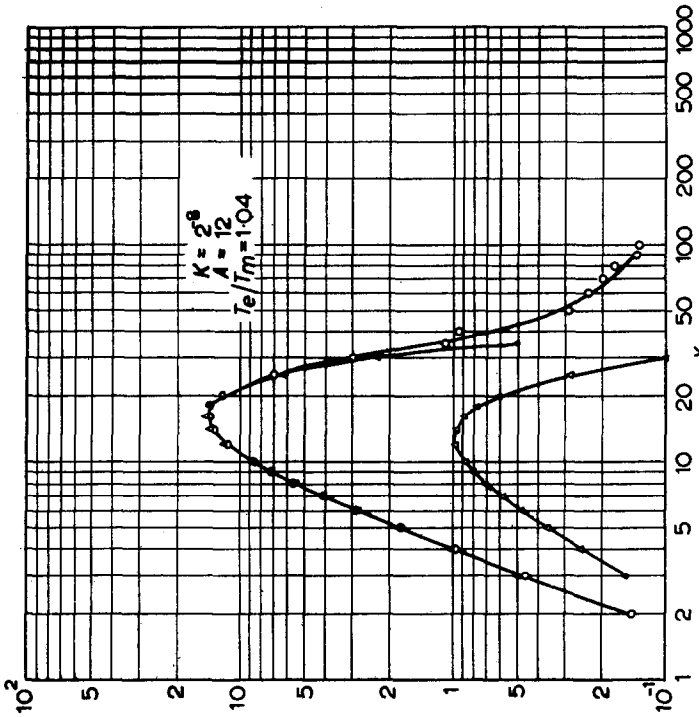


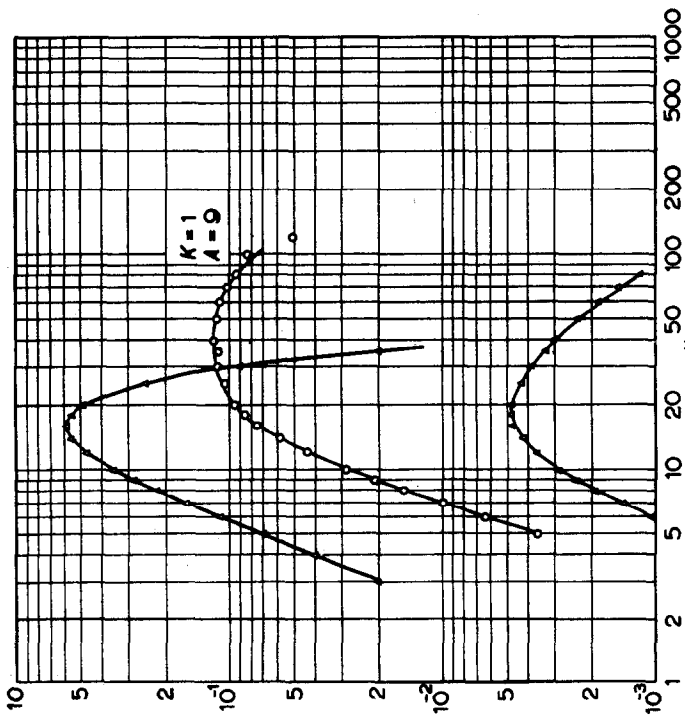
FIG. 5.

○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
△ (lower curve) actual number density = (actual flux)/X
□ ($1/v$ flux with same normalization)



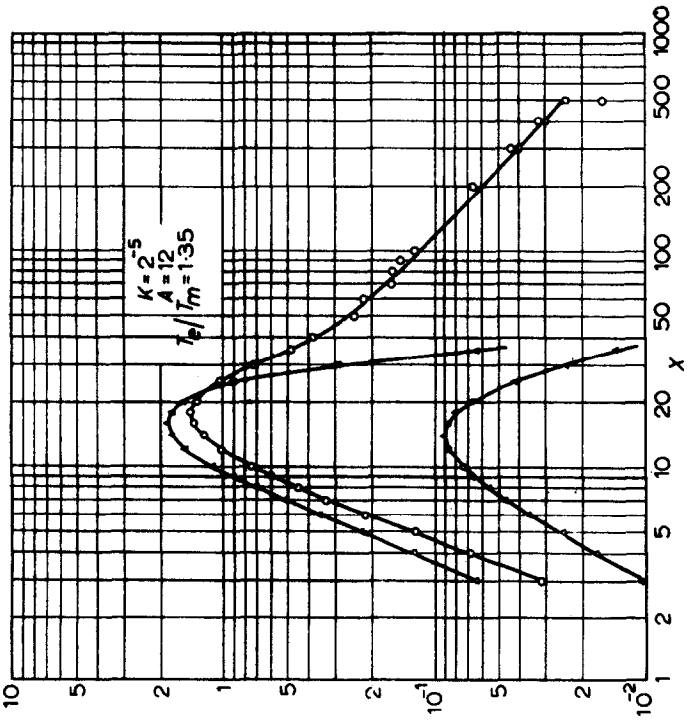
○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
△ (lower curve) actual number density = (actual flux)/X

FIG. 8.



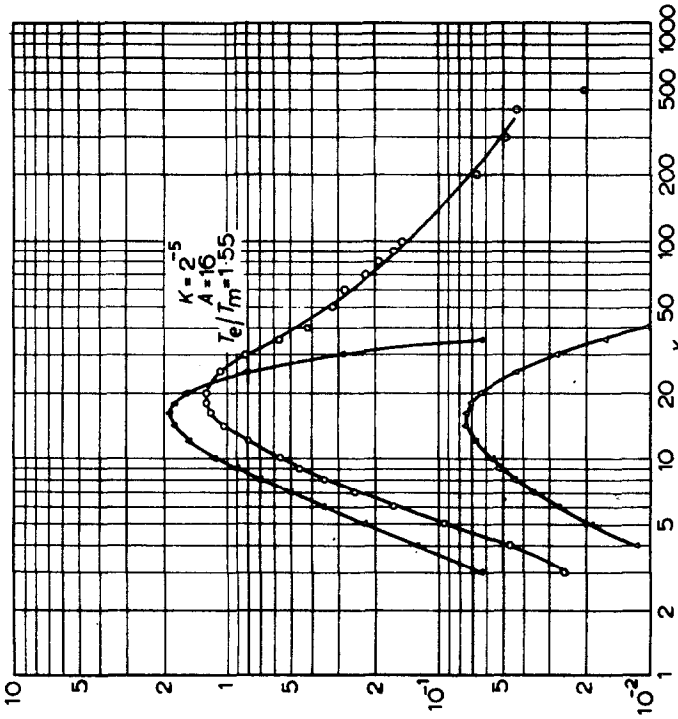
○ actua lflux
△ (upper curve) Maxwell-Boltzmann flux
△ (lower curve) actual number density = (actual flux)/X

FIG. 7.



○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
△ (lower curve) actual number density = (actual flux)/X
□ $(1/v)$ flux with same normalization

FIG. 9.



○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
△ (lower curve) actual number density = (actual flux)/X

FIG. 10.

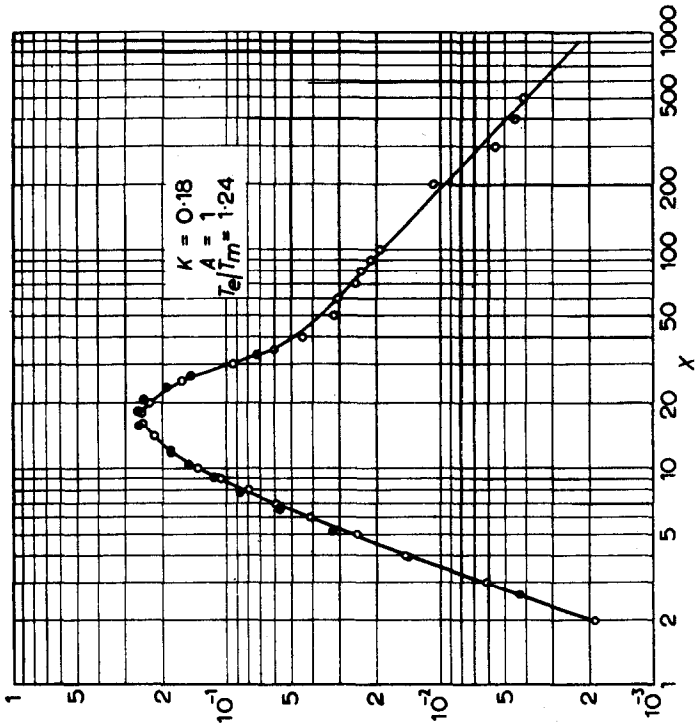


FIG. 12.
○ actual flux computed by Monte Carlo method
• actual flux computed by WIGNER and WILKINS

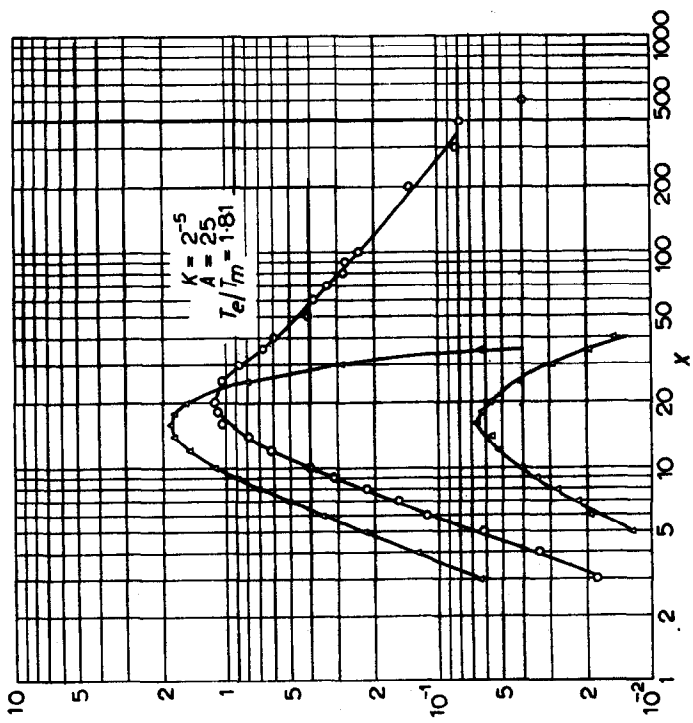


FIG. 11.
○ actual flux
△ (upper curve) Maxwell-Boltzmann flux
△ (lower curve) actual number density = actual flux)/X

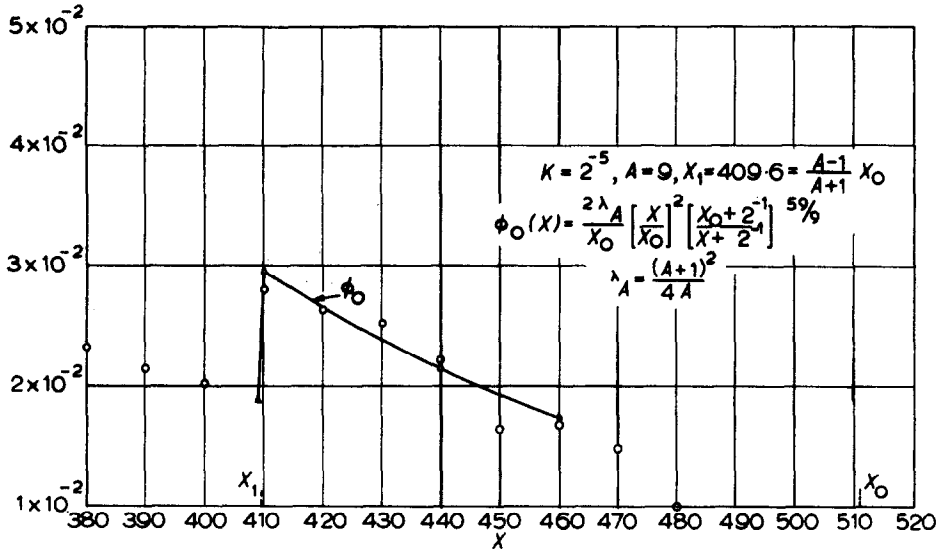

5×10⁻²
4×10⁻²
3×10⁻²
2×10⁻²
1×10⁻²
380 390 400 410 420 430 440 450 460 470 480 490 500 510 520
X
X₁
X₀
φ₀
K = 2⁻⁵, A = 9, X₁ = 409·6 = (A−1)/(A+1) X₀
φ₀(X) = (2λ_A/X₀) [X/X₀]² [(X₀+2⁻¹)/(X+2⁻¹)]⁵⁹/⁹
λ_A = (A+1)²/4A

FIG. 13.